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Fuses: Littelfuse MINI 58V (0997 series, 1 kA at 58 V DC) ONLY, in Littelfuse MINI FL1 holders (58 V DC, 22 A continuous). A 32 V MINI fuse fits the same holder: the silk says 58 V FUSES ONLY at every holder. Pull a fuse only with both packs out (D55).

High-side shunt R1 (1 mOhm) between J1 (from the MRBF load stud) and +BUS: the INA228's inputs stay inside −0.3..85 V of its ground (BUS−, the switched negative) whether the main switch is on or off. Positive current = into the packs (regeneration); discharge reads negative.

Kelvin taps (R1 pads 2, 3) at Bourns' measurement points; 10 Ohm in each input (TI SLYS021A 8.1.4); VBUS through 4.7 kOhm from the battery side.

Fit U1 (bare INA228; the I2C pull-ups are on the MCU carrier), or leave it off and wire an Adafruit 5832 (its 15 mOhm shunt removed, VBUS jumper open): VIN+ to TP2, VIN− to TP1, VBUS to TP3, its STEMMA QT lead to the MCU. Not both (both answer at 0x40).

J17 (JST GH 4-pin) pinout: 1 GND (BUS−), 2 3V3, 3 SDA, 4 SCL (the body makes this lead).

generated by hardware/fuse-board/generate.py

Gyropup body, task B5 (D60)

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File: fuse-board.kicad_sch

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